

NIKHIL GANPAT NAVGHADE

Senior Field Application Engineer | Automotive Radar | Radar SoCs | System Integration

Munich, Germany | +49 163 5454172 | nikhil.nawaghadej@gmail.com | <https://www.linkedin.com/in/nikhil-navghade/>

PROFESSIONAL SUMMARY

Senior Field Application / Radar Engineer with 12+ years of experience across automotive radar, radar SoCs, signal processing, embedded systems, system integration, validation and technical customer support. Combines deep radar algorithm knowledge with hands-on system debugging, RF validation and automation. Experienced in bridging application needs with engineering teams, troubleshooting HW/SW issues, supporting radar SoC integration, and translating complex radar-system behavior into actionable technical solutions.

CORE COMPETENCIES

- Field Application Engineering: Radar SoCs, system bring-up, technical troubleshooting, customer/application support
- Radar Systems: Automotive FMCW radar, radar signal chain, MIMO/DOA, detection, angle processing, radar performance
- System Integration & Validation: HW/SW integration, RF validation, automated testing, regression, root-cause analysis
- Radar Algorithms & DSP: Range/Doppler/angle processing, FFT, CFAR, beamforming, sidelobe suppression, noise estimation
- Automation & Software: Python, MATLAB, Embedded C, C, machine learning, edge AI, validation orchestration and data analysis
- RF / Lab Engineering: Radar evaluation boards, signal generators, spectrum analyzers, oscilloscopes, power supplies, power meters
- Architecture & Technical Leadership: Radar processing chains, software/module architecture, multi-core systems, memory optimization
- Technical Communication: Cross-functional engineering, customer issue analysis, workshops, demonstrations and R&D; coordination

PROFESSIONAL EXPERIENCE

NXP Semiconductors — System Integration & Validation Engineer

Mar 2026 – Present | Munich, Germany

- Develop automated system-level validation for automotive radar using an internal framework based mainly on Python, integrated with MATLAB and Java.
- Build validation workflows spanning radar configurations, temperature, power cycling, instrument settings and repeated execution, including runs with thousands of iterations.
- Use MATLAB for signal processing, measurement analysis, validation logic and plotting; use Java integration for radar-board and laboratory-instrument automation.
- Automate RF/lab equipment including R&S; SMA100B, R&S; FSW85, RTA2064, Agilent 33250A, power meters and LAN-controlled power supplies, depending on the test setup.
- Completed RX Filter Shaping validation and additional validation work covering Supply Ripple, Group/Chip-to-Chip Delay and Phase Rotation; currently working on Range-Doppler accuracy/separation.
- Extend the validation environment with reusable instrument-control interfaces, automated pass/fail evaluation and CVDB (PostgreSQL) result reporting for traceability and analysis.

Calterah GmbH — Field Application Engineer / Radar System Engineer

Sep 2024 – Feb 2026 | Munich, Germany

- Supported radar SoC application and system bring-up, debugging issues across acquisition, calibration, detection and angle-estimation functions.
- Investigated HW/SW integration problems and supported resolution of customer-reported radar-system issues across the processing pipeline and SoC features.
- Bridged application needs and China-based R&D;,, translating field feedback and technical requirements into engineering discussions and product actions.
- Delivered hands-on technical support, workshops and product demonstrations, including radar SoC demonstration at Electronica 2024.
- Worked on SoC/system topics including Secure Boot, OTP, Ethernet, Functional Safety and cybersecurity.

Fusionride GmbH — Senior Radar Signal Processing Engineer / Radar Expert

Feb 2022 – Sep 2024 | Germany

- Architected and developed radar signal-processing chains in MATLAB covering range, Doppler and angle processing, with MIMO and beamforming.
- Worked across radar algorithm architecture, software architecture, memory mapping, memory optimization, multi-core architecture, module integration and system debugging.
- Developed antenna/radar analysis tools for beam patterns, sidelobes and virtual-array evaluation, reducing analysis effort from approximately one week to one day.
- Contributed to development of a first 4x4 corner-radar prototype and subsequent 6x8 front-radar development; provided technical direction and module-level leadership.

Continental Automotive — Technical Specialist / Radar Algorithm Engineer

Jul 2018 – Feb 2022 | Bengaluru, India

- Developed radar signal-processing and algorithm modules for automotive FMCW radar systems.
- Implemented CFAR, elevation MIMO, sidelobe suppression and noise-estimation functions and supported system-level optimization.
- Developed radar processing modules across ARM Cortex-M, DSP and MATLAB environments with runtime/memory optimization and root-cause analysis.
- Contributed to production radar validation; reported functional coverage of 99.92% for a Gen5 radar validation activity.

Wavelet Technologies / Sci-com Software — Project Engineer – Radar DSP / Firmware

Jul 2015 – Jul 2018 | Pune, India

- Developed embedded firmware and DSP software for pulse / wind-profiler radar systems.
- Participated in radar development, debugging and product-lifecycle activities.

SELECTED NXP SYSTEM VALIDATION EXPERIENCE

- Supply Ripple Validation: automated RX/TX characterization using controlled ripple injection, ADC/Range-FFT analysis, spectrum-analyzer sideband measurements, automated pass/fail decisions and database reporting.
- Group / Chip-to-Chip Delay: synchronized ADC acquisition, FFT without windowing, frequency-domain cross-correlation, chirp averaging, phase extraction/unwrapping and linear phase fitting.
- Phase Rotation: fine chirp-to-chirp phase-step validation and INL/DNL-style step-error evaluation against defined limits.
- Validation Architecture: requirement-derived Excel inputs, layered orchestration, instrument-control interfaces, signal-processing analysis, regression execution and CVDB traceability.

TECHNICAL SKILLS

Radar: Automotive FMCW, pulse radar, radar SoCs, signal processing, MIMO, DOA, beamforming, CFAR, range/Doppler/angle processing, antenna analysis

Application / System: System integration, radar validation, RF validation, system bring-up, HW/SW debugging, root-cause analysis, cross-functional support

Programming: Python, MATLAB, Embedded C, C, Java

RF / Instruments: R&S; SMA100B, R&S; FSW85, RTA2064, Agilent 33250A, oscilloscopes, power meters, LAN-controlled power supplies

Architecture: Radar processing chains, software/module architecture, ARM Cortex-M, DSP, multi-core architecture, memory mapping/optimization

Data / Automation: Automation frameworks, test orchestration, automated measurement, pass/fail evaluation, CVDB, PostgreSQL

EDUCATION

- **M.E. Embedded Systems & VLSI** — Savitribai Phule Pune University, 2016 | CGPA 8.5/10
- **B.E. Electronics & Telecommunication** — Savitribai Phule Pune University, 2013 | CGPA 8.5/10

PUBLICATIONS & ACHIEVEMENTS

- IEEE Conference Paper: "Comparative study and implementation ... wind profiler radar", 2017.
- Developed a first 4x4 corner-radar prototype with real-time detections within one year at Fusionride.
- Received a monetary award for a tool that accelerated initial radar signal-processing bring-up and reduced engineering effort.

LANGUAGES

English | Hindi | Marathi | German